



*Master of Operations Research, Combinatorics and
Optimization*

Master Informatique / Master Mathématiques & Applications

QUANTIQUE

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This internship was done at the *Laboratory of Informatics of Grenoble* (LIG)
in the *CAPP - Calculi, Algorithms, Programs and Proofs* team.

Is under the supervision of *mehdi mhalla*.

Defended before a jury composed of:

↪ Van-Dat Cung
↪ Nadia Brauner

*This is a tremendously inspirational quote that
sets the tone of this course; truly, one of the
epigraphs of all time.*

— Jonathan Bingus

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CHAPTER 1

1.1. eee

d'après [\[Ji+22\]](#) et [\[Lin11\]](#) et [\[BBV20\]](#)

Bibliography

CHAPTER 1.1

- [BBV20] Matthias Beck, Benjamin Braun, and Andrés R. Vindas-Meléndez, “Decompositions of Ehrhart h^* -polynomials for rational polytopes”, 2020. [Online]. Available: <https://arxiv.org/abs/2006.10076>
- [Ji+22] Zhengfeng Ji, Anand Natarajan, Thomas Vidick, John Wright, and Henry Yuen, “MIP*=RE,” 2022. [Online]. Available: <https://arxiv.org/abs/2001.04383>
- [Lin11] Eva Linke, “Rational Ehrhart quasi-polynomials,” 2011. [Online]. Available: <https://arxiv.org/abs/1006.5612>